

Autonomous Research Outreach Agent

Technical Report: Workflow, Logic, and Architecture

ML Project — Autonomous Agent

April 13, 2026

Abstract

This report describes an end-to-end system for academic outreach: from a vague user goal through structured intent capture, execution planning, web discovery of faculty, relevance ranking, and personalized email drafting, with human supervision before sending and optional scheduled automation for replies and follow-ups. The implementation combines conversational LLM agents, a LangGraph-style orchestration graph, web search and scraping, composite relevance scoring, and a small persistence layer (SQLite) with Gmail integration.

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1 Introduction and objectives

The project implements an *agentic* pipeline that helps a student or researcher run a disciplined cold-email campaign toward faculty. The design emphasizes:

- **Structured intent** — slot-filling so vague goals become actionable constraints (domains, geography, timeline, profile).
- **Explicit plans** — a machine-readable execution plan (search, filters, email style, schedule) that the user can approve or refine.
- **Grounded outreach** — candidates are discovered from the open web and bibliographic APIs; emails reference real publications and lab context where possible.
- **Human oversight** — the first pass is supervised: the user confirms intent and plan, then reviews drafts before messages leave the system.
- **Gradual autonomy** — after approval and configuration, scheduled jobs can re-crawl for new leads, detect inbox replies, and send bounded follow-ups.

2 High-level architecture

Figure 1 summarizes how packages connect around shared `AgentState`; Figure 2 lists top-level folders in a non-overlapping grid. Figures 4–5 map the Typer CLI to control and data movement. Figure 3 groups the longer 14-step narrative into phases.

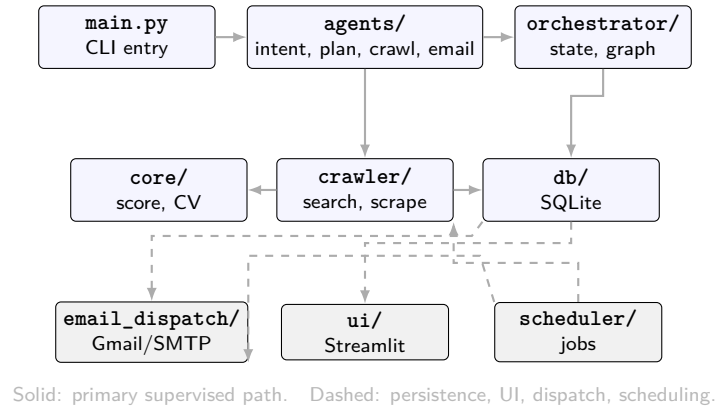


Figure 1: Repository modules and their main dependencies. `AgentState` lives in `orchestrator/`; the CLI path exercises agents, crawler, core scoring, and the database before optional dispatch and scheduled jobs.

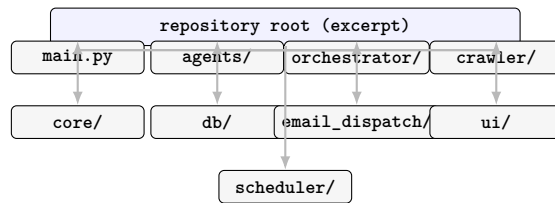
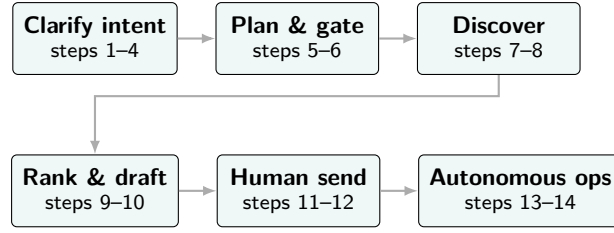


Figure 2: Top-level packages and entry script aligned on a grid (no overlap). Scripts under `tools/` and `scripts/` are omitted.

Two orchestration paths. The primary user flow in `main.py` runs a *sequential* supervised pipeline (five labeled CLI steps). An alternative `orchestrator/graph.py` compiles a **Lang-Graph** `StateGraph` with nodes `intent` → `planning` → `crawling` → `scoring` → `email_draft`



End-to-end workflow grouped by responsibility (Section 4).

Figure 3: Phased view of the 14-step workflow: from slot-filling through discovery, scoring, supervised sending, and optional scheduled re-crawl, replies, and follow-ups.



Figure 4: Supervised `main.py start` pipeline as labeled in the CLI (matches `main.py`).

→ `send` (emulated), using a memory checkpointer for stateful runs. Both paths share the same Pydantic/TypedDict state models (Figure 6).

3 Central state and data models

All agents read and write a shared `AgentState` (`orchestrator/state.py`), a `TypedDict` holding the keys below; Figure 7 shows them schematically.

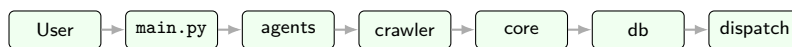
- `messages` — chat history for clarification and plan explanation.
- `intent` — serialized `UserIntent`.
- `plan` — serialized `ExecutionPlan`.
- `professors, shortlisted` — lists of `ProfessorProfile` dictionaries.
- `draft_emails` — `DraftEmail` records.
- `human_input, mode, session_id`, optional `error`.

3.1 User intent (`UserIntent`)

Progressively filled slots include outreach `goal` (enum, e.g. research internship), `research_domains`, `countries`, optional `target_institutions`, `timeline`, `user_name`, profile summary or `cv_path`, email tone, follow-up and volume preferences, and structured CV-derived fields (degree, institution, highlights, projects, etc.). Method `missing_required_slots()` drives which questions the intent agent asks next; `confirmed` gates transition to planning in graph routing. Figure 8 summarizes the loop.

3.2 Execution plan (`ExecutionPlan`)

Encodes `SearchStrategy` (keywords, country and institution filters), `FilteringCriteria` (minimum relevance score, require email, recency flags), `EmailStrategy` (personalization level, tone,



Simplified happy path: terminal I/O, LLM agents, web/API fetch, scoring, persistence, optional Gmail.

Figure 5: Simplified supervised-run data path (one direction of influence; loops for HITL omitted).

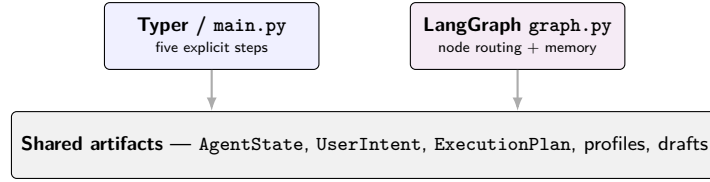


Figure 6: Dual orchestration entry points converging on the same typed state and persistence contracts.

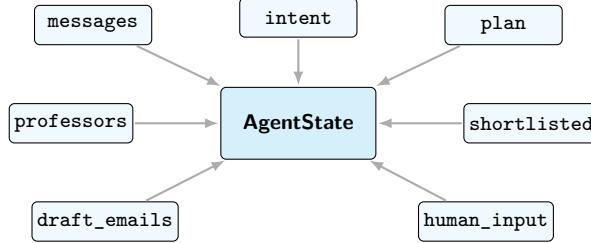


Figure 7: Shared `AgentState` hub and the main fields agents read or update (Section 3).

word count, CV attachment), `SendingSchedule` (emails per week, follow-up delay, max follow-ups), and `top_k_professors`. The planning LLM is instructed to preserve geographic strings exactly (e.g. “Punjab, India”) when copying filters.

3.3 Professor and email artifacts

`ProfessorProfile` stores identity, institution, keywords, publications, URLs, optional `accepting_students`, crawl metadata, and `relevance_score`. `DraftEmail` stores recipient, subject, body, status life-cycle (draft, approved, sent, replied, followed up), and timestamps. Figure 9 sketches the draft lifecycle relative to automation.

4 End-to-end workflow (14 steps)

The following maps your specified workflow to the implementation. Steps 1–6 are predominantly handled by `IntentAgent` and `PlanningAgent`; steps 7–10 by `CrawlerAgent`, `RelevanceScorer`, and `EmailAgent`; steps 11–12 by CLI review, the database, and Streamlit/Gmail dispatch; steps 13–14 by `scheduler/`. Figure 10 compresses the discovery stack; Figure 12 shows scheduled maintenance tasks.

1. User provides a vague high-level goal.

CLI prompt What do you want to do? or first `human_input` in the graph.

2. Agent extracts intent and identifies missing parameters.

`IntentAgent.extract_initial_intent` uses an LLM with a JSON schema prompt to populate `UserIntent`; `missing_required_slots()` lists gaps (domains, countries, timeline, name, profile/CV).

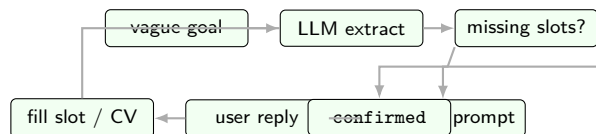


Figure 8: Slot-filling loop for `UserIntent`: questions repeat until required fields exist, then confirmation releases planning.

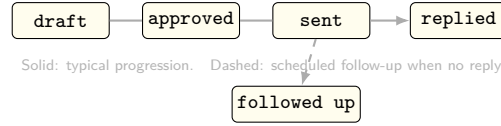


Figure 9: DraftEmail status progression (conceptual; exact flags depend on DB rows and scheduler).

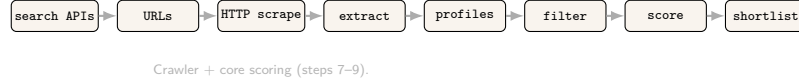


Figure 10: Discovery pipeline: queries and heuristics yield URLs; scraping and ProfileExtractor build ProfessorProfile rows, then filters and RelevanceScorer produce shortlisted.

3. Agent asks structured follow-up questions.

Template map `SLOT_QUESTIONS` in `intent_agent.py`; `fill_slot_from_reply` parses answers (including PDF CV via `CVParse`).

4. Agent builds structured state representation.

The running `UserIntent` model is serialized into `AgentState["intent"]`; optional persistence to `session_state.json` and `upsert_session_state` in the DB.

5. Agent generates and displays execution plan.

`PlanningAgent.generate_plan` produces `ExecutionPlan` JSON via LLM, with fallback `_default_plan`; `display_plan` and a plain-language `PLAN_EXPLANATION_PROMPT` explain it to the user.

6. User approves plan.

In `run_cli`, responses `yes/no` trigger `confirm` or `refine_plan`. In `LangGraph`, `evaluate_plan` routes to crawling only when `plan.confirmed` is true.

7. Agent crawls web sources.

`CrawlerAgent.run`: web search (`search_professors`), URL heuristics (`is_likely_faculty_url`), Semantic Scholar batch queries, optional snowball expansion, then `scrape_batch` and `ProfileExtractor.batch_extract`.

8. Agent extracts candidate professor data.

Structured `ProfessorProfile` instances with email, keywords, publications, etc.; optional filter requiring a discoverable email address.

9. Agent ranks candidates by relevance.

`RelevanceScorer.shortlist`: geographic pre-filter when countries are specific; composite score

$$s = \sum_i w_i s_i, \quad w = (0.30, 0.20, 0.15, 0.25, 0.10)$$

for keyword overlap, LLM “embedding” slot (student–lab fit score), publication recency, geographic match, and student-acceptance signal; then threshold by `min_relevance_score` and take top- K . Figure 11 shows the relative weights.

10. Agent drafts personalized emails.

`EmailAgent.generate_email`: salutation sanitization, student context from `build_email_internship_cont` JSON generation prompt with length targets, QC/retry, optional critique pass, and template `_fallback_email` if quality checks fail.

11. User supervises first execution.

`main.py` loops per draft: `display_email`, user chooses `approve/skip`; approved rows go to `db.save_draft_email`. Streamlit python `main.py` review supports further review and sending.

12. Emails are sent.

Gmail/SMTP clients under `email_dispatch/`; sending updates persisted in the database (implementation details depend on UI and OAuth setup).

13. Agent tracks replies and sends follow-ups.

`ReplyMonitor.scan_and_update_replies` matches inbox senders to sent rows and marks replied. `FollowupEngine.process_pending_followups` sends polite follow-ups after `followup_days` without a reply, respecting max follow-up policy encoded in intent/`plan`.

14. After approval, system transitions to autonomous mode.

`AutonomousScheduler` (`APScheduler`) schedules `re_crawl_job` (discover new professors, skip already-contacted emails), `check_replies_job`, and `send_followups_job`. This phase assumes Gmail and DB configuration and user consent to automated actions.

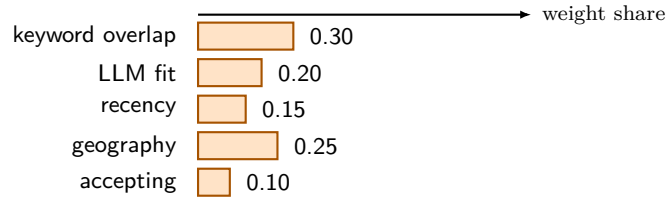


Figure 11: Composite relevance score $s = \sum_i w_i s_i$: bar length is w_i (Section 4, step 9).

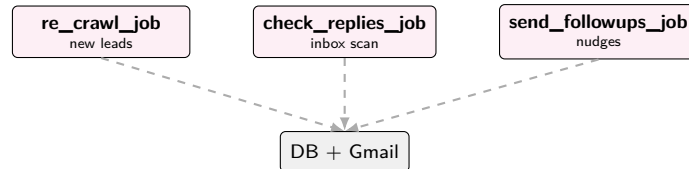


Figure 12: Autonomous scheduler jobs (`APScheduler`) and their shared sinks after the user enables automation (Section 4, step 14).

5 Control flow and routing logic

5.1 LangGraph conditional edges

Figure 13 sketches the graph; the bullets spell out the routing predicates.

- After `intent`: if `intent.confirmed`, go to `planning`; else loop `intent`.
- After `planning`: if `plan.confirmed`, go to `crawling`; else loop `planning`.
- After `crawling`: unconditional edge to `scoring`.
- After `scoring`: if `shortlisted` is empty, `END`; else `email_draft`.
- After `email_draft`: edge to `send` (CLI graph uses `send_emulated` for simulation), then `END`.

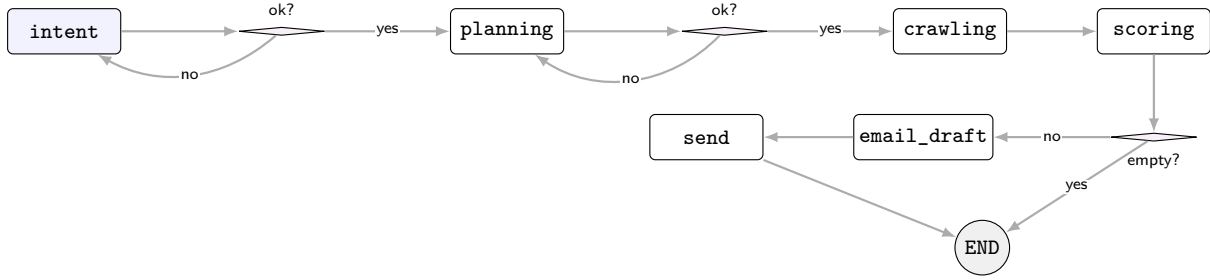


Figure 13: LangGraph-style routing (Section 5). Diamonds abbreviate `intent.confirmed`, `plan.confirmed`, and `shortlisted empty`; `send` may be emulated in CLI runs.

5.2 Supervised CLI loop (`run_supervised_loop`)

Streams graph events and pauses for terminal input when intent or plan is not yet confirmed, injecting the next `human_input` and resuming—a lightweight human-in-the-loop pattern without full LangGraph interrupt nodes. Figure 14 contrasts the two confirmation gates.

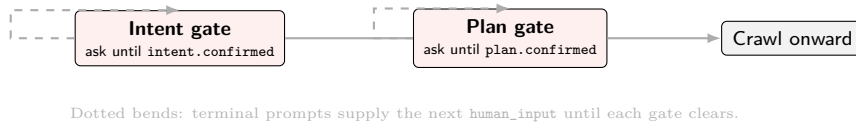


Figure 14: Human-in-the-loop pattern in the supervised CLI before expensive crawling.

6 Key design decisions

Slot-filling vs. free chat

Clarification is template-driven and JSON-parseable, reducing ambiguity before expensive crawl/send steps.

Hybrid relevance

Lexical overlap plus an LLM fit score balances interpretability with semantic matching when CV signals exist.

Email quality gates

Salutation guards, keyword sanitization, word-count targets, and a critique pass mitigate malformed or generic outreach.

Persistence boundaries

Session intent/plan in JSON; drafts and sent mail in SQLite; OAuth tokens external to this report.

7 Documented live run (exact session log)

This section attaches one **real** supervised run: `python3 main.py start` under `scripts/run_agent_logged.py`, tee'd to `run_logs/agent_20260413T140344Z.log`. The **framed listing below is the full transcript** as captured (no bullet summaries): every line from environment warnings through the final prompt, in order.

The file `transcript_ascii.txt` (upload next to this `.tex` on Overleaf) matches that session after ANSI strip; for pdf_{La}TeX only, Unicode is mapped to ASCII (`[ok]`, `[!]`, box-drawing to `+`, `|`, `-`, etc.). The raw UTF-8 Rich log remains `run_logs/agent_20260413T140344Z.log` in the repo.

Identifiers in the log. Natural-language replies, institutions, and addresses appear exactly as in the terminal output.

7.1 Session provenance

- **Raw log:** `run_logs/agent_20260413T140344Z.log`
- **Transcript file (Overleaf):** `transcript_ascii.txt`
- **Launcher:** `scripts/run_agent_logged.py` **Pipeline:** `main.py start` (interactive Prompt.ask; not a scripted demo).

7.2 Exact console transcript

The box spans pages as needed; contents are *verbatim* from `transcript_ascii.txt` (lines 1–end, including the initial urllib/LibreSSL warning).

Complete session log — exact transcript

```
/Users/harsh/Library/Python/3.9/lib/python/site-packages/urllib3/__init__.py:35:
  NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module
    is compiled with 'LibreSSL 2.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
warnings.warn(
+-----+
| Welcome to the Autonomous Research Outreach Agent |
| Let's find the perfect professors and draft personalized emails. |
+-----+
[ok] Database initialized.
Database initialized.

=== Step 1/5: Intent Clarification ===
+-----+
| Research Outreach Agent |
| I'll help you set up a personalized academic outreach campaign. |
+-----+

What do you want to do?: I want a research internship in Game Theory in Punjab, India

Analyzing your request...

I need a few more details. (3 questions remaining)

> What is your target timeline? (e.g. Summer 2025, Fall 2025, immediately)
: Summer 2026

> What is your full name? (for personalizing the emails)
: Harsh Modi

> Please briefly describe your background: your degree, year, key skills, and
notable projects or publications. (Or type 'cv' to provide a CV file path)
: '/Users/harsh/Desktop/ML/MLProjectAutonomousAgent/HarshModi.pdf'
Parsing CV: /Users/harsh/Desktop/ML/MLProjectAutonomousAgent/HarshModi.pdf
[ok] CV parsed successfully.

Generating confirmation summary...

+----- Confirmation -----+
| To confirm, here's a summary of your structured outreach plan: |
| |
| * Your goal is to secure a research internship, specifically in the domain |
| of Game Theory. |
```



```
| * You're interested in opportunities located in Punjab, India. |
| * Your background is in Computer Science and Engineering, with a strong |
| foundation in AI, NLP, and formal verification, and you're currently a |
| student at the Indian Institute of Technology, Ropar. |
| * You have a strong portfolio of projects, including stock price prediction, |
| formal verification systems, and blockchain integration plugins. |
| * Your research interests include AI, NLP, formal verification, time series |
| prediction, and blockchain. |
| * You're looking to apply for internships during Summer 2026. |
| * You'd like to send 10 outreach emails per week, with a follow-up email |
| after 7 days if you don't receive a response. |
| * You prefer a professional tone for your emails. |
| * Your CV is available at the specified path and highlights your |
| achievements, skills, and experience. |
| |
| Does this look correct? (yes to confirm / no to make changes) |
```

-----+
Your answer: yes

[ok] Intent confirmed! Moving to planning...

=== Step 2/5: Execution Planning ===

Generating execution plan...

-----+
| Generated Execution Plan |

Search Strategy

-----+
| Field | Value |

-----+
Keywords	Game Theory, AI, NLP, Formal Verification, Time Series
	Prediction, Blockchain, Machine Learning, Conversational
	AI, Stock Price Prediction
Countries	Punjab, India
Top-K Professors	2
-----+

Email Strategy

-----+
| Field | Value |

-----+
Personalization	high
Tone	professional
Word Count Target	220
Ref. Publications	[ok]
Attach CV	[ok]
-----+

Sending Schedule

-----+
| Field | Value |

-----+
Emails/Week	10
Follow-up After	7 days
Max Follow-ups	2
-----+

Getting plain-language explanation...

-----+ What this means -----+
| Here's a simple explanation of the outreach execution plan: |
| |

```
| * **Search approach**: The plan will search for professors using keywords |
| like Game Theory, AI, and Machine Learning, and filter results to only |
| include professors from Punjab, India. |
| * **Professor selection**: Professors will be selected based on criteria |
| such as: |
| * Recent publications |
| * Active lab membership |
| * Relevance to the search keywords (with a minimum score of 0.4) |
| * Availability of an email address |
| * **Email strategy**: Emails will be: |
| * Highly personalized |
| * Written in a professional tone |
| * Around 220 words long |
| * Include a CV attachment and reference to relevant publications |
| * Sent to a maximum of 2 professors at a time, with follow-ups scheduled 7 |
| days apart. |
+-----+
```

Approve this plan? (yes / no + describe change)

: yes

[ok] Plan approved! Starting crawl...

=== Step 3/5: Crawling for Professors ===

```
+-----+
| Web Crawling Mode |
| Searching for professors matching your criteria... |
+-----+
```

Step 1/3: Running web searches...

Determining top institutions in Punjab, India...

Found regional institutions: IIT Ropar, Thapar University, PEC Chandigarh, NIT Jalandhar, GNDU Amritsar

Running 38 search queries...

```
(1/38) Game Theory professor Punjab, India research internship
(2/38) site:.ac.in Game Theory faculty professor Punjab, India
(3/38) Game Theory professor IIT Ropar research
(4/38) Game Theory professor Thapar University research
(5/38) Game Theory professor PEC Chandigarh research
(6/38) Game Theory professor NIT Jalandhar research
(7/38) Game Theory professor GNDU Amritsar research
(8/38) AI professor Punjab, India research internship
(9/38) site:.ac.in AI faculty professor Punjab, India
(10/38) AI professor IIT Ropar research
(11/38) AI professor Thapar University research
(12/38) AI professor PEC Chandigarh research
(13/38) AI professor NIT Jalandhar research
(14/38) AI professor GNDU Amritsar research
(15/38) NLP professor Punjab, India research internship
(16/38) site:.ac.in NLP faculty professor Punjab, India
(17/38) NLP professor IIT Ropar research
(18/38) NLP professor Thapar University research
(19/38) NLP professor PEC Chandigarh research
(20/38) NLP professor NIT Jalandhar research
(21/38) NLP professor GNDU Amritsar research
(22/38) Formal Verification professor Punjab, India research internship
(23/38) site:.ac.in Formal Verification faculty professor Punjab, India
(24/38) Formal Verification professor IIT Ropar research
(25/38) Formal Verification professor Thapar University research
(26/38) Formal Verification professor PEC Chandigarh research
(27/38) Formal Verification professor NIT Jalandhar research
```

(28/38) Formal Verification professor GNDU Amritsar research
 (29/38) Time Series Prediction professor Punjab, India research internship
 (30/38) site:.ac.in Time Series Prediction faculty professor Punjab, India
 (31/38) Time Series Prediction professor IIT Ropar research
 (32/38) Time Series Prediction professor Thapar University research
 (33/38) Time Series Prediction professor PEC Chandigarh research
 (34/38) Time Series Prediction professor NIT Jalandhar research
 (35/38) Time Series Prediction professor GNDU Amritsar research
 (36/38) Game Theory research group lab Punjab, India PhD students
 (37/38) AI research group lab Punjab, India PhD students
 (38/38) NLP research group lab Punjab, India PhD students
 [ok] Found 311 unique candidate URLs
 [ok] 50 faculty URLs identified for scraping

Step 2/3: Querying Semantic Scholar...

Searching Semantic Scholar...

Keyword (1/8): Game Theory

Keyword (2/8): AI

Keyword (3/8): NLP

Keyword (4/8): Formal Verification

Keyword (5/8): Time Series Prediction

Keyword (6/8): Blockchain

Semantic Scholar search error: 429 Client Error: for url:

<https://api.semanticscholar.org/graph/v1/author/search?query=Blockchain&limit=15&fields=name%2Caffiliations%2CpaperCount%2CcitationCount%2ChIndex%2Cpapers.title%2Cpapers.year%2Cpapers.venue>

Keyword (7/8): Machine Learning

Keyword (8/8): Conversational AI

[ok] Found 48 unique authors

Snowball Expansion: Fetching active co-authors...

[ok] Snowball expansion discovered 2 relevant co-authors

Step 3/3: Scraping 50 pages...

(1/50) Scraping: https://cup.edu.in/summer_res_internship...

? Summer/Research Internship | (no email)

(2/50) Scraping:

<https://www.isb.edu/faculty-and-research/faculty-directory/abhinav-upp...>

? abhinav uppal | abhinav_uppal@isb.edu

(3/50) Scraping:

https://www.punjabiversity.ac.in/Departments/Profiles/18/Profile_P1...

? (name not found) | (no email)

(4/50) Scraping:

https://dtu.ac.in/modules/faculty_profile_new/faculty_index.php?uname=...

Fetch error (https://dtu.ac.in/modules/faculty_profile_new/faculty_index...):

HTTPSConnectionPool(host='dtu.ac.in', port=443): Max retries exceeded with url: /modules/faculty_profile_new/faculty_index.php?uname=jamkhongam (Caused by ConnectTimeoutError(<urllib3.connection.HTTPSConnection object at 0x115f803a0>, 'Connection to dtu.ac.in timed out. (connect timeout=15)'))

? (name not found) | (no email)

(5/50) Scraping:

https://punjabiversity.ac.in/Departments/Profiles/78/Profile_P78X4C...

? (name not found) | (no email)

(6/50) Scraping: <https://iiitvadodara.ac.in/faculty/ag001.php...>

? IIIT | antriksh_goswami@iiitvadodara.ac.in

(7/50) Scraping: <https://www.iitrpr.ac.in/math/?p=2893...>

? Dr. Arvind Kumar Gupta | ak Gupta@iitrpr.ac.in

(8/50) Scraping:

<https://punjabiversity.ac.in/pages/Department.aspx?dsenc=54...>

? Punjabi University, Patiala | (no email)

(9/50) Scraping:

<https://www.iitj.ac.in/People/Profile/ccf498a7-2d28-4410-a814-aecff198...>

? S Srivatsa Srinivas | (no email)
 (10/50) Scraping: <https://www.iitrpr.ac.in/maths/?p=2848...>
 ? Dr. Arti Pandey | arti@iitrpr.ac.in
 (11/50) Scraping:
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=swapnil.dhamal...>
 ? Research Highlights | swapnil.dhamal@iitrpr.ac.in
 (12/50) Scraping: https://www.iitrpr.ac.in/maths/?page_id=773...
 ? Ph.D. | (no email)
 (13/50) Scraping:
<https://iitrpr.irins.org/faculty/index/Department+of+Computer+Science+...>
 ? IIT Ropar | (no email)
 (14/50) Scraping: https://cse.iitrpr.ac.in/?page_id=83...
 ? Research Areas | (no email)
 (15/50) Scraping: <https://www.iitrpr.ac.in/AK/PEOPLE.html...>
 ? People | poonamr@iitrpr.ac.in
 (16/50) Scraping:
<https://scholar.google.com/citations?user=067LPdgAAAAJ&hl=en...>
 Skipping direct scrape of Google Scholar (avoids 403, no emails anyway).
 ? (name not found) | (no email)
 (17/50) Scraping:
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=shwetajain%40i...>
 ? Research Highlights | shwetajain@iitrpr.ac.in
 (18/50) Scraping: <http://www.cs.iit.edu/~kapoor/...>
 ? Sanjiv Kapoor | (no email)
 (19/50) Scraping: <https://eced.thapar.edu/facultydetails/MTIxNA==...>
 ? (name not found) | sanjay.sharma@thapar.edu
 (20/50) Scraping: <https://tslas.thapar.edu/facultymaster/97...>
 ? AMIT PRAKASH JHA | amit.prakash@thapar.edu
 (21/50) Scraping: <https://som.thapar.edu/facultydetails/MTQ1NA==...>
 ? School of Mathematics | tina.verma1@thapar.edu
 (22/50) Scraping: <https://eced.thapar.edu/facultydetails/MTU3NW==...>
 ? (name not found) | debayani.ghosh@thapar.edu
 (23/50) Scraping: <https://som.thapar.edu/facultydetails/NjM2...>
 ? School of Mathematics | amitkumar@thapar.edu
 (24/50) Scraping:
<https://pec.ac.in/sites/default/files/pdf/Merged-IDP-2016.pdf...>
 ? (name not found) | (no email)
 (25/50) Scraping: <https://pec.ac.in/mathematics/faculty...>
 ? Faculty | (no email)
 (26/50) Scraping: <https://curin.chitkara.edu.in/profile/dr-bhisham-sharma/...>
 ? BHISHAM SHARMA Associate Professor | (no email)
 (27/50) Scraping: <https://pec.ac.in/mathematics/faculty/ankit-kumar...>
 ? Ankit Kumar | (no email)
 (28/50) Scraping:
<https://www.iimamritsar.ac.in/faculty-and-research/faculty-user-profil...>
 ? Faculty Directory | pavneets@iimamritsar.ac.in
 (29/50) Scraping:
<https://ptu.ac.in/wp-content/uploads/2024/05/updated-list-of-Ph.D.-awa...>
 ? (name not found) | (no email)
 (30/50) Scraping:
<https://pec.ac.in/sites/default/files/2025-11/mechanical-engineering-s...>
 ? (name not found) | (no email)
 (31/50) Scraping:
https://pec.ac.in/sites/default/files/2024-11/Mathematics_computing_sy...
 ? (name not found) | (no email)
 (32/50) Scraping:
<https://nitj.irins.org/faculty/index/Department+of+Mathematics...>
 ? NIT Jalandhar | (no email)
 (33/50) Scraping: https://nitj.ac.in/research/research_publications.html...
 ? Research Publications | (no email)
 (34/50) Scraping:
https://www.nitj.ac.in/nitj_files/links/CCSP_Brochure_ECE_Tentative_7_...

? (name not found) | (no email)
 (35/50) Scraping:
<https://rndshowcase.iiit.ac.in/tto2022/main1.php?centername=Machine%20...>
 ? Research | (no email)
 (36/50) Scraping:
https://digital.gndu.ac.in/Final_View_Faculty_Proforma.aspx?mid=354456...
 ? GURU NANAK DEV UNIVERSITY | mlsingh.ece@gndu.ac.in
 (37/50) Scraping: <https://digital.gndu.ac.in/pdf/resume/2015101199.pdf...>
 ? (name not found) | (no email)
 (38/50) Scraping:
https://digital.gndu.ac.in/Final_View_Faculty_Proforma.aspx?mid=354456...
 ? GURU NANAK DEV UNIVERSITY | aman.phy@gndu.ac.in
 (39/50) Scraping:
<https://www.iitj.ac.in/school-of-artificial-intelligence-data-science/...>
 ? Internship Programs | (no email)
 (40/50) Scraping:
https://konnifel.com/professor/public_internship/a1kzL013K3BLNnI2LytqV...
 ? AI Research Assistantship | (no email)
 (41/50) Scraping: <https://www.daiict.ac.in/summer-research-internship-dau...>
 ? logo | (no email)
 (42/50) Scraping:
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=Arnksh%40iitrpr...>
 ? Research Highlights | Arnksh@iitrpr.ac.in
 (43/50) Scraping:
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=mayuresh%40iit...>
 ? Research Highlights | mayuresh@iitrpr.ac.in
 (44/50) Scraping: <https://pec.ac.in/faculty/cse...>
 ? Faculty | (no email)
 (45/50) Scraping:
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=awaneesh%40iit...>
 ? Research Highlights | awaneesh@iitrpr.ac.in
 (46/50) Scraping:
https://digital.gndu.ac.in/Final_View_Faculty_Proforma.aspx?mid=354456...
 ? GURU NANAK DEV UNIVERSITY | kuldeep.ece@gndu.ac.in
 (47/50) Scraping:
https://punjabiversity.ac.in/Departments/Profiles/18/Profile_P18X8A...
 ? (name not found) | (no email)
 (48/50) Scraping:
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=ashwani.sharma...>
 ? Research Highlights | ashwani.sharma@iitrpr.ac.in
 (49/50) Scraping:
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=reddy%40iitrpr...>
 ? Research Highlights | reddy@iitrpr.ac.in
 (50/50) Scraping:
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=puneet.kumar%4...>
 ? Research Highlights | puneet.kumar@iitrpr.ac.in

Extracting professor profiles...

(1/50) Extracting from https://cup.edu.in/summer_res_internship...
 [ok] Summer/Research Internship | Central University of Punjab
 (2/50) Extracting from
<https://www.isb.edu/faculty-and-research/faculty-directory/a...>
 [ok] Abhinav Uppal | Indian School of Business
 (3/50) Extracting from
<https://www.punjabiversity.ac.in/Departments/Profiles/18/...>
 [x] Skipped (no name found)
 (4/50) Extracting from
https://dtu.ac.in/modules/faculty_profile_new/faculty_index...
 [x] Skipped (no name found)
 (5/50) Extracting from
<https://punjabiversity.ac.in/Departments/Profiles/78/Prof...>
 [x] Skipped (no name found)

(6/50) Extracting from <https://iiitvadodara.ac.in/faculty/ag001.php...>
 [ok] Antriksh Goswami | Indian Institute of Information Technology Vadodara

(7/50) Extracting from <https://www.iitrpr.ac.in/math/?p=2893...>
 [ok] Arvind Kumar Gupta | Indian Institute of Technology Ropar

(8/50) Extracting from
<https://punjabiversity.ac.in/pages/Department.aspx?dsenc=...>
 [ok] Punjabi University, Patiala | Punjabi University

(9/50) Extracting from
<https://www.iitj.ac.in/People/Profile/ccf498a7-2d28-4410-a81...>
 [ok] S Srivatsa Srinivas | Indian Institute of Technology Jodhpur

(10/50) Extracting from <https://www.iitrpr.ac.in/math/?p=2848...>
 [ok] Dr. Arti Pandey | Indian Institute of Technology Ropar

(11/50) Extracting from
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=swap...>
 [ok] Swapnil Vilas Dhamal | IIT Ropar

(12/50) Extracting from https://www.iitrpr.ac.in/math/?page_id=773...
 [ok] Ph.D. | IIT Ropar

(13/50) Extracting from
<https://iitrpr.irins.org/faculty/index/Department+of+Compute...>
 [ok] Nitin Auluck | IIT Ropar

(14/50) Extracting from https://cse.iitrpr.ac.in/?page_id=83...
 [ok] Research Areas | IIT Ropar

(15/50) Extracting from <https://www.iitrpr.ac.in/AK/PEOPLE.html...>
 [ok] Arvind Kumar Gupta | Indian Institute of Technology Ropar

(16/50) Extracting from
<https://scholar.google.com/citations?user=067LPdgAAAAJ&hl=en...>
 [x] Skipped (no name found)

(17/50) Extracting from
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=shwe...>
 [ok] Shweta Jain | IIT Ropar

(18/50) Extracting from <http://www.cs.iit.edu/~kapoor/...>
 [ok] Sanjiv Kapoor | Illinois Institute of Technology

(19/50) Extracting from <https://eced.thapar.edu/facultydetails/MTIxNA==...>
 [ok] Sanjay Sharma | Thapar Institute of Engineering and Technology

(20/50) Extracting from <https://tslas.thapar.edu/facultymaster/97...>
 [ok] Amit Prakash Jha | Thapar Institute of Engineering and Technology

(21/50) Extracting from <https://som.thapar.edu/facultydetails/MTQ1NA==...>
 [ok] School of Mathematics | THAPAR

(22/50) Extracting from <https://eced.thapar.edu/facultydetails/MTU3Nw==...>
 [ok] Debayani Ghosh | Thapar Institute of Engineering and Technology

(23/50) Extracting from <https://som.thapar.edu/facultydetails/NjM2...>
 [ok] Amit Kumar | Thapar Institute of Engineering and Technology

(24/50) Extracting from
<https://pec.ac.in/sites/default/files/pdf/Merged-IDP-2016.pd...>
 [x] Skipped (no name found)

(25/50) Extracting from <https://pec.ac.in/mathematics/faculty...>
 [ok] Ankit Kumar | Punjab Engineering College (Deemed to be University), Chandigarh

(26/50) Extracting from
<https://curin.chitkara.edu.in/profile/dr-bhisham-sharma/...>
 [ok] Bhisham Sharma | Chitkara University

(27/50) Extracting from <https://pec.ac.in/mathematics/faculty/ankit-kumar...>
 [ok] Ankit Kumar | Punjab Engineering College, (Deemed to be University), Chandigarh

(28/50) Extracting from
<https://www.iimamritsar.ac.in/faculty-and-research/faculty-u...>
 [ok] Pavneet Singh | IIM Amritsar

(29/50) Extracting from
<https://ptu.ac.in/wp-content/uploads/2024/05/updated-list-of...>
 [x] Skipped (no name found)

(30/50) Extracting from
<https://pec.ac.in/sites/default/files/2025-11/mechanical-eng...>

[x] Skipped (no name found)
 (31/50) Extracting from
https://pec.ac.in/sites/default/files/2024-11/Mathematics_co...
 [x] Skipped (no name found)
 (32/50) Extracting from
<https://nitj.irins.org/faculty/index/Department+of+Mathemati...>
 [ok] Dr Jaspal Aujla | NIT Jalandhar
 (33/50) Extracting from
https://nitj.ac.in/research/research_publications.html...
 [ok] Research Publications | AC
 (34/50) Extracting from
https://www.nitj.ac.in/nitj_files/links/CCSP_Brochure_ECE_Te...
 [x] Skipped (no name found)
 (35/50) Extracting from
<https://rndshowcase.iiit.ac.in/tto2022/main1.php?centername=...>
 [ok] Research | IIIT
 (36/50) Extracting from
https://digital.gndu.ac.in/Final_View_Faculty_Proforma.aspx?...
 [ok] Dr. Maninder Lal Singh | Guru Nanak Dev University
 (37/50) Extracting from
<https://digital.gndu.ac.in/pdf/resume/2015101199.pdf...>
 [x] Skipped (no name found)
 (38/50) Extracting from
https://digital.gndu.ac.in/Final_View_Faculty_Proforma.aspx?...
 [ok] Dr. Aman Mahajan | Guru Nanak Dev University
 (39/50) Extracting from
<https://www.iitj.ac.in/school-of-artificial-intelligence-dat...>
 [ok] Internship Programs | IIT Jodhpur
 (40/50) Extracting from
https://konnifel.com/professor/public_internship/a1kzL013K3B...
 [ok] AI Research Assistantship | Central University of Punjab, Bathinda
 (41/50) Extracting from
<https://www.daiict.ac.in/summer-research-internship-dau...>
 [ok] Abhishek Jindal | Dhirubhai Ambani Institute of Information and Communication Technology
 (42/50) Extracting from
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=Arnk...>
 [ok] Arun Kumar Sharma | IIT Ropar
 (43/50) Extracting from
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=mayu...>
 [ok] Dr. M. S. Pardeshi | IIT Ropar
 (44/50) Extracting from <https://pec.ac.in/faculty/cse...>
 [ok] Akashdeep Kaur | Punjab Engineering College (Deemed to be University), Chandigarh
 (45/50) Extracting from
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=awan...>
 [ok] Awaneesh Kumar Yadav | Indian Institute of Technology Ropar
 (46/50) Extracting from
https://digital.gndu.ac.in/Final_View_Faculty_Proforma.aspx?...
 [ok] Dr. Kuldeep Singh | Guru Nanak Dev University
 (47/50) Extracting from
<https://punjabiversity.ac.in/Departments/Profiles/18/Prof...>
 [x] Skipped (no name found)
 (48/50) Extracting from
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=ashw...>
 [ok] Ashwani Sharma | IIT Ropar
 (49/50) Extracting from
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=redd...>
 [ok] C.C. Reddy | Indian Institute of Technology Ropar
 (50/50) Extracting from
<https://www.iitrpr.ac.in/saide/profile.faculty.php?mail=pune...>
 [ok] Puneet Kumar | Indian Institute of Technology (IIT) Ropar

```

Processing 50 Semantic Scholar authors...

[ok] Extracted 75 unique professor profiles (14 duplicates removed)

Email filter: kept 25/75 professors with known emails

[ok] Crawl complete. 25 professor profiles ready for scoring.

=== Step 4/5: Scoring & Shortlisting (25 candidates) ===
Geographic filter: dropped 3 professors not in Punjab, India

Scoring 22 professors...
[ok] Scoring complete.

Top 2 Shortlisted Professors
+-----+-----+-----+-----+-----+-----+-----+
| # | Name | Instituti? | Domains | Score | Email | Accepting? |
+-----+-----+-----+-----+-----+-----+-----+
| 1 | Shweta | IIT Ropar | Game | 0.490 | shwetajai? | ? |
| | Jain | | Theory, | | |
| | | Machine | | |
| | | Learning, | | |
| | | Rein | | |
| 2 | Awaneesh | Indian | Applied | 0.488 | awaneesh@? | ? |
| | Kumar | Institute | Cryptogra? | | |
| | Yadav | of Techn | Post | | |
| | | Quantum | | |
+-----+-----+-----+-----+-----+-----+

[ok] 2 professors shortlisted.

=== Step 5/5: Generating & Reviewing Emails ===

? Generating 2 personalized emails...

(1/2) Shweta Jain --- IIT Ropar
[!] Email QC failed for Shweta Jain; retrying once...
[!] Expanding short email for Shweta Jain...
[ok] Subject: Research Internship Inquiry --- Shweta Jain
(2/2) Awaneesh Kumar Yadav --- Indian Institute of Technology Ropar
[!] Email QC failed for Awaneesh Kumar Yadav; retrying once...
[!] Expanding short email for Awaneesh Kumar Yadav...
[ok] Subject: Research Internship Inquiry --- Awaneesh Kumar Yadav

[ok] Generated 2 draft emails.

--- Email 1/2 ---
+-----+-----+-----+-----+-----+-----+-----+
| To: shwetajain@iitrpr.ac.in |
| Subject: Research Internship Inquiry --- Shweta Jain |
| ----- |
| Dear Professor Jain, |
| |
| I am excited to apply for a summer research internship under your guidance |
| at IIT Ropar, where I can leverage my strong foundation in AI, NLP, and |
| formal verification to explore real-world problems. Your work on Exploring |
| and Mitigating Gender Bias in Book Recommender Systems and EQBAL-RS: |
| Mitigating popularity bias in recommender systems resonates with my research |
| interests in Game Theory. I am impressed by your research on Efficient |
| Algorithms For Fair Clustering with a new fairness notion, which aligns with |
| my skills and interests. |
| |
| In my previous projects, I developed AI-powered formal verification systems |

```



```

| and built conversational AI agents with advanced reasoning capabilities. I |
| am confident that my skills will enable me to contribute to your research |
| group. I am particularly drawn to your work because it addresses critical |
| issues in recommender systems, and I believe my strong academic background |
| and research experience make me a strong fit for your team. I look forward |
| to discussing my application and how I can contribute to your research |
| endeavors, specifically focusing on mitigating biases in recommender |
| systems. |
| |
| Sincerely, |
| Harsh Modi |
| ----- |
| Note: (Critiqued & Revised by LLM Reviewer) |
+-----+

```

Action [approve/skip/quit] (approve): approve
[ok] Draft saved to database.

--- Email 2/2 ---

```

+- Draft Email ? Awaneesh Kumar Yadav | Indian Institute of Technology Ropar ---+
| To: awaneesh@iitrpr.ac.in |
| Subject: Research Internship Inquiry --- Awaneesh Kumar Yadav |
| ----- |
| Dear Professor Yadav, |
| |
| I am Harsh Modi, a Bachelor of Technology student in Computer Science and |
| Engineering at the Indian Institute of Technology, Ropar, with a strong |
| foundation in AI, NLP, and formal verification. Your work in Applied |
| Cryptography, Post Quantum Security, and AI security for 5G/6G has been a |
| significant influence on my research interests. I am particularly drawn to |
| the applications of Game Theory in these fields, as evident in your |
| publications on Post Quantum Cryptography and Quantum Key Distribution. |
| |
| With experience in developing AI-powered formal verification systems and |
| building conversational AI agents, I am confident that my skills align with |
| your research group's focus. I am excited about the opportunity to |
| contribute to and learn from your projects, specifically those related to |
| 5G/6G Security and Post Quantum Security. I believe that an internship under |
| your guidance would be invaluable in exploring these areas further. |
| |
| I would appreciate the chance to discuss my application and how I can |
| contribute to your research group. I look forward to the opportunity to work |
| with you and explore the applications of Game Theory in Post Quantum |
| Security. |
| |
| Sincerely, |
| Harsh Modi |
| ----- |
| Note: (Critiqued & Revised by LLM Reviewer) |
+-----+

```

Action [approve/skip/quit] (approve): approve
[ok] Draft saved to database.

[ok] Review complete! 2 approved, 0 skipped.

Tip: Run `python3 main.py review` to launch the UI and send approved emails.

[ok] Agent workflow completed.

8 Limitations and operational notes

- Web scraping depends on page structure and rate limits; some faculty pages yield incomplete profiles.
- Search API keys and LLM backends must be configured via environment variables (`config.py`); the stack may use Groq or other providers as wired in code.
- Autonomous mode sends real email; users should verify legal, institutional, and etiquette constraints before enabling schedulers.

9 Conclusion

The project implements a coherent pipeline from natural-language goals to structured state, approved plans, grounded discovery, ranked candidates, and supervised email generation, with a path toward scheduled re-crawling, reply tracking, and follow-ups. The 14-step workflow above matches the intended product narrative and aligns with the main classes and modules in the repository. Section 7 embeds the full supervised-session log verbatim (Game Theory / Punjab, India run) in a framed, page-breaking listing.